discrete vehicles, each with its genes disciplined and ordered by the prospect of a shared bottleneck of sperms or eggs funnelling them into the future. But this is not a fact to be taken for granted. It is a fact to be questioned and wondered at in its own right. Why did genes come together into large vehicles, each with a single genetic exit route? Why did genes choose to gang up and make large bodies for themselves to live in? In *The Extended Phenotype* I attempt to work out an answer to this difficult problem. Here I can sketch only a part of that answer—although, as might be expected after seven years, I can also now take it a little further.

I shall divide the question up into three. Why did genes gang up in cells? Why did cells gang up in many-celled bodies? And why did bodies adopt what I shall call a 'bottlenecked' life cycle?

First then, why did genes gang up in cells? Why did those ancient replicators give up the cavalier freedom of the primeval soup and take to swarming in huge colonies? Why do they cooperate? We can see part of the answer by looking at how modern DNA molecules cooperate in the chemical factories that are living cells. DNA molecules make proteins. Proteins work as enzymes, catalysing particular chemical reactions. Often a single chemical reaction is not sufficient to synthesize a useful end-product. In a human pharmaceutical factory the synthesis of a useful chemical needs a production line. The starting chemical cannot be transformed directly into the desired end-product. A series of intermediates must be synthesized in strict sequence. Much of a research chemist's ingenuity goes into devising pathways of feasible intermediates between starting chemicals and desired end-products. In the same way single enzymes in a living cell usually cannot, on their own, achieve the synthesis of a useful end-product from a given starting chemical. A whole set of enzymes is necessary, one to catalyse the transformation of the raw material into the first intermediate, another to catalyse the transformation of the first intermediate into the second, and so on.

Each of these enzymes is made by one gene. If a sequence of six enzymes is needed for a particular synthetic pathway, all six genes for making them must be present. Now it is quite likely that there are two alternative pathways for arriving at that same end-product, each needing six different enzymes, and with nothing to choose between the two of them. This kind of thing happens in chemical factories. Which pathway